CYBER SECURITY

TOOLS & CAREER OVERVIEW

Task 5 — Cyber Security Internship Report

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Topics Covered: Nmap | Wireshark | Antivirus | Firewalls | Career Paths | Certifications | Ethics | Personal Learning Summary

4	7	6	5
Security tools covered	Career roles explained	Certifications listed	Ethics principles

1. Nmap — Network Mapper

Nmap (Network Mapper) is a free, open-source security tool used by network administrators and ethical hackers worldwide to discover devices on a network, find open ports, detect running services, and identify operating systems. Created in 1997 by Gordon Lyon (Fyodor), it is one of the most essential tools in any cyber security professional's toolkit.

Full Name	Network Mapper (Nmap)
Developer	Gordon Lyon (Fyodor) — Free & Open Source
Platform	Windows, Linux, macOS — Command-line + GUI (Zenmap)

Website	nmap.org — free to download
Primary Purpose	Network discovery, port scanning, security auditing
Key Command	nmap -sV 192.168.1.1 — scans host, detects services & versions
Legal Note	Only scan networks you OWN or have written permission to test

What Nmap Does — Simple Explanation:

Imagine your office building has 65,535 doors (ports). Nmap checks each door and reports: which doors are open, what service is behind each door, and which operating system the building runs. This helps security teams find doors that should be locked but are not — before an attacker finds them first.

Real-World Example — Nmap Scan Output

```
$ nmap -sV 192.168.1.1
Starting Nmap 7.94 ( https://nmap.org )
Nmap scan report for 192.168.1.1
Host is up (0.0012s latency)
PORT STATE SERVICE VERSION
22/tcp open ssh OpenSSH 8.9
80/tcp open http Apache httpd 2.4.52
443/tcp open https Apache httpd 2.4.52 3306/tcp open
mysql MySQL 8.0.32  SECURITY RISK if public!
8080/tcp open http-proxy Squid http proxy 5.7
Nmap done: 1 IP address (1 host up) scanned in 12.43 seconds
```

Analysis: Port 3306 (MySQL database) is open to the public internet — this is a critical security finding. A database should never be directly accessible from the internet. Nmap helps identify exactly this kind of dangerous misconfiguration.

2. Wireshark — Packet Analyser

Wireshark is a free, open-source network protocol analyser that captures every data packet travelling on a network in real time. It lets you look inside each packet and see exactly what data is being sent between computers. Think of it as a microscope for your internet traffic — everything becomes visible.

Full Name	Wireshark Network Analyser
Developer	Wireshark Foundation — Free & Open Source

Platform	Windows, Linux, macOS — graphical interface
Website	wireshark.org — free to download
Primary Use	Live packet capture, protocol analysis, security investigation
Key Feature	Colour-coded traffic, filter by protocol, export captures for analysis
Security Use	Detects unencrypted passwords (HTTP), suspicious traffic, malware beacons
Learning Use	Best tool to visually understand how TCP/IP, DNS, HTTP protocols work

What Wireshark Reveals — Simple Explanation:

Every time your computer visits a website, sends an email, or loads an app, data is broken into small pieces called "packets" and sent across the network. Wireshark captures all these packets and lets you read them. On HTTP (not HTTPS) connections, this means you can see usernames and passwords in plain text — which is exactly why all modern websites must use HTTPS.

Real-World Example — Wireshark Packet Capture

No.	Time	Source	Destination	Protocol	Info
1	0.000	192.168.1.5	8.8.8.8	DNS	Standard query A google.com
2	0.002	8.8.8.8	192.168.1.5	DNS	Response: 142.250.76.46
3	0.004	192.168.1.5	142.250.76.46	TCP	SYN Seq=0 Win=65535
4	0.006	142.250.76.46	192.168.1.5	TCP	SYN-ACK Seq=0 Ack=1
5	0.008	192.168.1.5	142.250.76.46	HTTPS	Application Data (encrypted)
6	**0.015**	**192.168.1.5**	**10.0.0.100**	**HTTP**	**POST /login.php (PLAINTEXT!)**
7	0.016	10.0.0.100	192.168.1.5	HTTP	HTTP/1.1 200 OK (login success)

Row 6 (highlighted red): An HTTP POST to /login.php — this is a login form submission over unencrypted HTTP. Wireshark can read the username and password in plain text from this packet. This is why every login page must use HTTPS.

3. Antivirus Software

Antivirus software is a program that detects, prevents, and removes malicious software (malware) from computers. Modern antivirus uses two main detection methods: signature-based detection (matching known threats from a database) and behaviour-based detection (spotting suspicious actions even from new, unknown threats).

How it works	Scans files against a database of known malware signatures — updated daily
Behaviour analysis	Monitors program actions in real-time — detects new/unknown malware
Real-time protection	Intercepts and scans files before they open or execute
What it detects	Viruses, trojans, ransomware, spyware, adware, worms, rootkits
Popular options	Windows Defender (free, built-in), Kaspersky, Norton, Bitdefender, Malwarebytes
Limitation	Cannot stop all threats — phishing emails that steal passwords bypass antivirus
Best practice	Keep antivirus updated and run full system scans weekly

Types of Malware Antivirus Defends Against:

Malware Type	What it Does	Real Example
Virus	Attaches to files, spreads when files are shared	ILOVEYOU virus (2000) — $10B damage
Ransomware	Encrypts your files and demands payment to unlock them	WannaCry (2017) — NHS hospitals shut down
Trojan	Disguised as useful software, secretly gives attacker access	Zeus Trojan — stole banking credentials
Spyware	Silently monitors your computer activity and keystrokes	Pegasus (spyware targeting journalists)
Worm	Self-replicating — spreads across network without user action	SQL Slammer worm — crashed internet in 2003
Adware	Shows unwanted ads and redirects browser searches	Superfish (installed by Lenovo in 2015)

4. Firewalls

A firewall is a network security system that monitors and controls all incoming and outgoing network traffic based on a set of security rules. It acts as a barrier (a wall) between your trusted internal network and the untrusted internet. Everything that enters or leaves your network must pass through the firewall.

Simple Analogy	A security guard at the door — checks every visitor before letting them in
Packet Filtering	Examines each packet's source, destination, and port — allows or blocks based on rules
Stateful Inspection	Tracks connection state and context — smarter than simple packet filtering
Application Layer	Deep inspection of HTTP, FTP, DNS traffic — can block specific apps or websites
Software Firewalls	Windows Defender Firewall, iptables (Linux) — run on individual computers
Hardware Firewalls	Physical network devices — Cisco ASA, Palo Alto, pfSense, Fortinet
Next-Gen Firewalls	Combines firewall + IDS/IPS + app control + deep packet inspection in one device

How a Firewall Decision Works:

Rule #	Source	Destination	Port	Action	Reason
1	Any	192.168.1.10	22 (SSH)	ALLOW	Admin remote access
2	Any	192.168.1.10	80 (HTTP)	ALLOW	Web server traffic
3	Any	192.168.1.10	443 (HTTPS)	ALLOW	Secure web traffic
4	**Any**	**192.168.1.10**	**3306 (MySQL)**	**BLOCK**	**DB must not be public**
5	10.0.0.0/24	Any	Any	ALLOW	Internal network OK
6	**Any**	**Any**	**Any**	**BLOCK**	**Default deny all**

Rule #4 and #6 (highlighted red) show the most important firewall principles: block the database from public access, and deny everything that is not explicitly allowed.

5. Cyber Security Career Paths

Cyber security is one of the fastest-growing career fields globally. With over 3.5 million unfilled positions worldwide (Cybersecurity Ventures, 2024), demand far exceeds supply. In India, the sector is growing at 15% annually with strong salaries at every level. Here are the main career paths:

Career Role	Level	Salary (India)	Key Responsibilities	Skills Needed
SOC Analyst (Security Ops Center)	Entry Level	**■4–8 LPA**	Monitors security dashboards 24/7, responds to alerts, investigates incidents	SIEM tools, network analysis, incident response
Ethical Hacker / Pen Tester	Mid Level	**■8–18 LPA**	Legally hacks company systems to find vulnerabilities before real attackers do	Kali Linux, Metasploit, scripting, web app testing
Digital Forensics Analyst	Mid Level	**■6–14 LPA**	Investigates cyber crimes, recovers deleted files and digital evidence for courts	Autopsy, FTK, evidence handling, report writing
Security Engineer	Senior Level	**■15–30 LPA**	Designs and builds security systems, firewalls, IDS/IPS, SIEM infrastructure	Network security, cloud security, architecture design
Malware Analyst	Mid Level	**■8–16 LPA**	Reverse engineers malware samples to understand attack methods and create defences	Assembly language, reverse engineering, sandboxing
CISO (Chief Info Sec Officer)	Executive	**■40–80 LPA**	Heads all security for an organisation, creates security strategy, reports to board	Leadership, risk management, compliance, 10+ years exp.
Bug Bounty Hunter	Freelance	**Variable**	Finds security bugs in websites and apps, earns rewards from companies like Google	Web hacking, ethical hacking skills, report writing

Career Tip: Start with CompTIA Security+ → get an entry-level SOC job → build skills → specialize in penetration testing or cloud security for top salaries.

6. Cyber Security Certifications Roadmap

Certifications validate your skills and are often required for hiring. The right certification path depends on your career goal. Here is the recommended progression from beginner to expert:

Certificate	Level	Approx. Cost	Focus	Best For
CompTIA Security+	Beginner	$370 USD	Core security concepts, threats, cryptography	First certification — get this first
eJPT (eLearnSecurity)	Beginner	$200 USD	Hands-on penetration testing basics	Before OSCP — great practical start
CEH (EC-Council)	Intermediate	$1,199 USD	Ethical hacking methodologies, tools	Recognised by many Indian companies
CompTIA CySA+	Intermediate	$370 USD	Threat detection, SOC analyst skills	For SOC analyst career path
OSCP (Offensive Security)	Advanced	$1,499 USD	24-hour practical hacking exam — very hard	Penetration testing gold standard
CISSP	Expert	$749 USD	Security management, risk, architecture	Senior roles, CISO career path

7. Ethics in Cyber Security

Ethics is the foundation of cyber security. The difference between a security professional and a criminal is not technical skill — it is permission and intent. Every cyber security professional must follow a strict code of ethics.

Written Permission is Mandatory

NEVER scan, test, access, or probe any system you do not own or have explicit written permission to test. Even well-intentioned unauthorised testing is illegal. Always get a signed scope-of-work document before any security testing.

Responsible Disclosure

If you discover a vulnerability in a company's website or system, report it to them privately and give them reasonable time to fix it (typically 90 days) before making it public. Many companies have official Bug Bounty programs for exactly this purpose.

Protect User Privacy

During security work, you may encounter personal data — usernames, emails, financial records. Never access data beyond what is needed, never store it unnecessarily, and never share it with anyone. User privacy is always paramount.

Know the Legal Boundaries

In India, unauthorised computer access is a criminal offence under IT Act 2000 (Section 66) and can result in imprisonment up to 3 years and a fine. "Just testing" or "I found the vulnerability" are not legal defences without prior permission.

Professional Integrity

Security professionals are entrusted with extremely sensitive information about an organisation's weaknesses. Betraying this trust — by sharing findings, exploiting vulnerabilities, or working for adversaries — destroys careers and leads to prosecution.

Golden Rule: If you did not get written permission, DO NOT TEST IT. Ethical hackers are only ethical because they have permission.

8. Personal Learning Summary

1. Understanding the Attacker's View

Learning Nmap and Wireshark gave me a completely new perspective — I now understand how an attacker "sees" a network. Every open port is a potential entry point, and every unencrypted packet is readable data. This changed how I think about designing secure systems.

2. Tools are Only as Good as Their User

Nmap, Wireshark, and other tools are just instruments. A hammer can build a house or break a window. The same tools used by defenders are used by attackers. What separates them is knowledge, intent, and permission.

3. Career Possibilities Opened Up

Before this task, I thought "cyber security" was one job. Now I see it is an entire ecosystem — SOC analysts, ethical hackers, forensics investigators, security engineers, and more. Each path requires different strengths, and there is a role that fits any background.

4. Ethics is Non-Negotiable

The most important lesson was about ethics. Permission is everything in this field. I will always obtain proper authorisation before conducting any security testing, and will report vulnerabilities responsibly rather than exploiting them.

5. Continuous Learning is Essential

The cyber security field changes faster than almost any other. New vulnerabilities, tools, and attack techniques emerge daily. The professionals who succeed are those who commit to lifelong learning through practice labs, CTF challenges, and certifications.

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